

NIHARIKA KANDEPU

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SUMMARY

AI/ML and Cloud Engineer with hands-on experience in **deep learning, computer vision, and blockchain-enabled healthcare systems**. Proficient in **Python, TensorFlow, Keras, AWS, Docker, and REST APIs**, with experience in **model training, evaluation, and deployment** of scalable ML and web applications. Strong foundation in **cloud infrastructure, MLOps fundamentals, and secure application development**.

PROFESSIONAL EXPERIENCE

- Northern Arizona University | Teaching Assistant-Python | AZ, USA****2024 – Present**
 - Assist in teaching Python to 120+ students, covering topics such as OOP, data structures, file handling, exceptions, and algorithm design.
 - Led lab sessions and code reviews on NumPy, Pandas, matplotlib, and REST API usage, improved assignment quality and on-time submissions by 65%.
 - Delivered personalized debugging help and project guidance, reducing student error rates by 35% and boosting overall technical comprehension.
- Cloud Computing Intern, KaaShiv InfoTech – AWS | Vijayawada, India****January 2024 – March 2024**
 - Completed a 3-month cloud computing internship under Microsoft MVP mentorship, focusing on AWS services, virtualization, and scalable deployment strategies.
 - Built and deployed secure cloud applications on AWS (EC2, S3, IAM, VPC) with a focus on scalability, CI/CD readiness, and high availability.
 - Applied real-time project scenarios to deepen proficiency in AWS infrastructure, improving deployment efficiency and system reliability.

PROJECTS

- An Exploratory Study on Healthcare Applications for Remote Patients****IEEE AIROSIP Conference, Aug 2023**
 - Designed a remote healthcare application using AI, VR, blockchain, and wearable data integration, enabling accessible care for remote patients.
 - Conducted interviews with 20+ users and experts; analyzed qualitative data to identify adoption barriers like privacy risks and digital literacy gaps.
 - Developed secure, scalable app architecture with chatbot support and remote monitoring, reducing patient travel needs and hospital visits.
- Advancing Eye Health with Convolution Neural Networks Using CNN and TensorFlow-Keras**
 - Built and trained a CNN model using TensorFlow to classify retinal images into cataract, glaucoma, diabetic retinopathy, and normal classes with 83.5% validation accuracy.
 - Trained CNN models on 2K+ retinal images using data augmentation techniques in TensorFlow/Keras, improving model generalization and accuracy.
 - Optimized and prepared models for deployment using Adam optimizer, early stopping, and checkpointing, ensuring efficient inference performance.
- Blockchain-Based Document Authentication System**
 - Built a decentralized document authentication Dap using Ethereum smart contracts and IPFS for secure, tamper-proof document storage and verification.
 - Designed a user-friendly frontend with React and Tailwind CSS, integrating MetaMask for wallet-based authentication and Hardhat for local blockchain development.
 - Implemented end-to-end workflow for document upload, SHA-256 hash generation, blockchain registration, and on-chain verification via smart contract logic.
- Care Connect – Centralized Online Health Record System.**
 - Developed a centralized electronic health record system using React, Angular, Node.js, Django, HTML5, CSS3, and JavaScript for seamless data access across providers.
 - Implemented secure, HIPAA-compliant architecture with encryption and authentication, enabling cloud deployment of scalable systems with real-time APIs suitable for ML inference integration.
 - Designed intuitive user interfaces for patients and professionals to manage appointments, view records, and facilitate communication, improving care coordination and operational efficiency.

TECHNICAL SKILLS

Programming Languages & Frameworks: Python, JavaScript, React, Node.js, Django, HTML5, CSS3
Machine Learning & Deep Learning: TensorFlow, Keras, CNNs, Computer Vision
Data Augmentation, Model Training, Model Evaluation, Inference
MLOps Fundamentals: Model Deployment, Training Pipelines | Containerization (Docker), Deployment Readiness
Cloud & DevOps: AWS (EC2, S3, IAM, VPC), GCP | Cloud Deployment, Scalable Systems | Docker, Kubernetes, CI/CD
Blockchain & Web3: Ethereum, Solidity (Smart Contracts) | Hardhat, IPFS, MetaMask, SHA-256
Databases & APIs: REST APIs, Real-time APIs | Data Handling, Backend Integration
Tools & Platforms: Pandas, NumPy | Git, Visual Studio Code
Soft Skills: Problem-Solving, Leadership, Team Collaboration, Adaptability

EDUCATION

- Masters of Information Technology****Aug 2024 – May 2026**
Northern Arizona University | Flagstaff, AZ, USA
(CGPA – 3.7/4)
- Bachelor of Technology in Computer Science****Aug 2020 – May 2024**
KL University | vaddeswaram, India
(CGPA – 8.55/10)

ACHIEVEMENTS

- IEEE AIROSIP 2023 Publication
- AWS Certified Cloud Practitioner